Show that $\mathbb{C}(\phi)$ is a transcendental extension of $\mathbb{C}$. Derive from this that $\mathbb{C}(\phi)$ is uncountably dimensional and obtain a contradiction.

2.4. Ideals

A **left ideal** of an algebra A is a subspace $I \subseteq A$ such that $aI \subseteq I$ for all $a \in A$. Similarly, a **right ideal** of an algebra A is a subspace $I \subseteq A$ such that $Ia \subseteq I$ for all $a \in A$. A **two-sided ideal** is a subspace that is both a left and a right ideal.

Left ideals are the same as subrepresentations of the regular representation A. Right ideals are the same as subrepresentations of the regular representation of the opposite algebra A^{op}.

Below are some examples of ideals:

- If A is any algebra, 0 and A are two-sided ideals. An algebra $A \neq 0$ is called **simple** if 0 and A are its only two-sided ideals.
- If $\phi : A \to B$ is a homomorphism of algebras, then $\ker \phi$ is a two-sided ideal of A.
- If S is any subset of an algebra A, then the two-sided ideal **generated** by S is denoted by $\langle S \rangle$ and is the span of elements of the form asb, where $a, b \in A$ and $s \in S$. Similarly, we can define $\langle S \rangle_\ell = \mathrm{span}\{as\}$ and $\langle S \rangle_r = \mathrm{span}\{sb\}$, the left, respectively right, ideal generated by S.

Problem 2.4.1. A **maximal** ideal in a ring A is an ideal $I \neq A$ such that any strictly larger ideal coincides with A. (This definition is made for left, right, or two-sided ideals.) Show that any unital ring has a maximal left, right, and two-sided ideal. (Hint: Use Zorn's lemma.)

2.5. Quotients

Let A be an algebra and let I be a two-sided ideal in A. Then A/I is the set of (additive) cosets of I. Let $\pi : A \to A/I$ be the quotient map. We can define multiplication in A/I by $\pi(a) \cdot \pi(b) := \pi(ab)$.